National University of Computer and Emerging Sciences, Lahore Campus

Course:	Electronic Devices & Circuits	Course Code:	EE225
Program:	BS(Electrical Engineering)	Semester:	Fall 2016
Duration:	60 Minutes	Total Marks:	40
Paper Date:	11-Nov-16	Weight	15%
Section:	ALL	Page(s):	2
Exam:	Midterm-2	Roll No:	

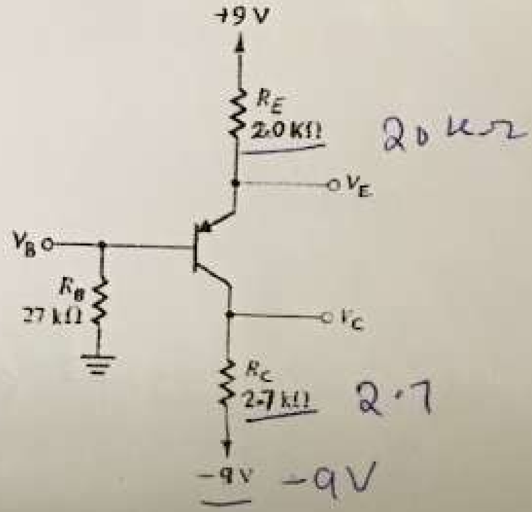
Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

In the circuit shown, the transistor has $\beta = 30$.

a) Find the values of V_B, V_E and V_C?

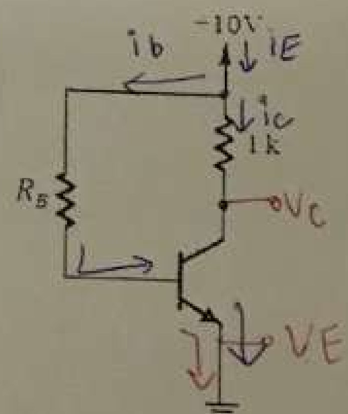
b) Find the largest value to which R_C can be raised while the transistor remains in the active mode. – 21.07



Question # 2 [10 marks]

For the circuit shown in figure, select a value of R_B so that the transistor saturates with an overdrive factor of 10. The BJT is specified to have a minimum β of 30. What is the value of forced β achieved? 3

97.5



V

E B C

Question # 3 [15 marks]

A common emitter amplifier utilizes a BJT with $\beta = 200$ and $V_A = 200$ V. It is biased at $I_C = 0.5$ mA and has a collector resistance $R_C = 10$ kΩ.

a) Draw and label the small signal equivalent circuit using Hybrid-π model including source and load resistance.

b) Find R_{in} , R_o and A_{vo}?

c) If the amplifier is fed with a signal source having a resistance of 10 kΩ and a load resistance R_L of 10 kΩ is connected to the output terminal, Find A_v and G_v?

d) If a peak-to-peak output ac voltage of 1V is measured at the collector, what ac input voltage and current must be associated with the base?